EELU
الجامعة المصرية للتعلم الإلكتروني الأهلية
THE EGYPTIAN E-LEARNING UNIVERSITY

Fall Semester

2025-2026

Information Technology

IT438

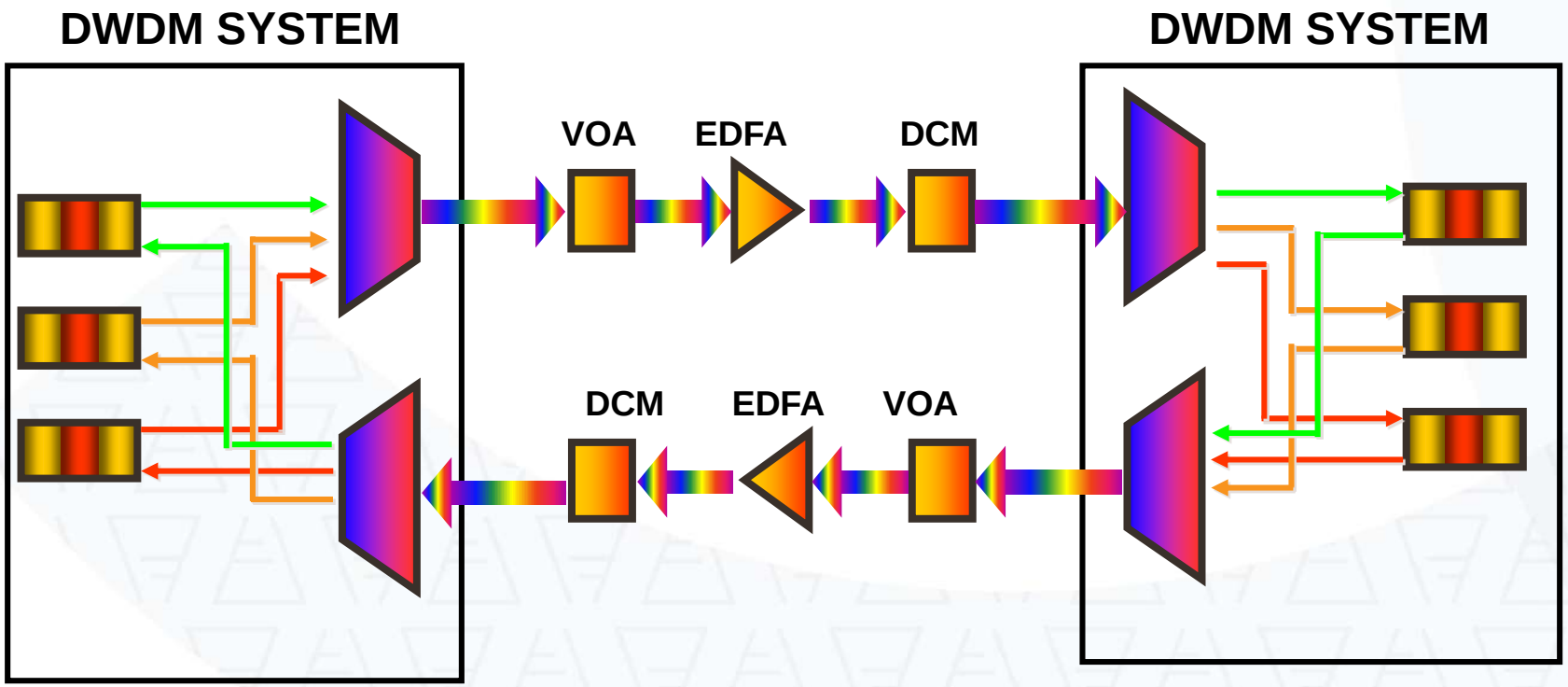
Communication Technology

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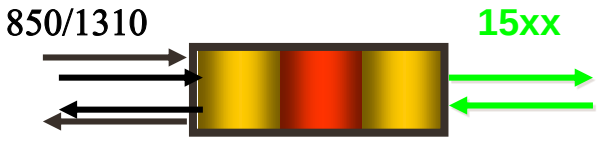
Acknowledgement: *This presentation contains some figures and text from Data Communications and Networks book by W. Stallings*

Typical DWDM Network



Transponders

Transponders



Transponder

- Critical component in DWDM systems, typically placed between the Terminal side equipment (e.g., routers or switches) and the DWDM system.

Its primary functions are:

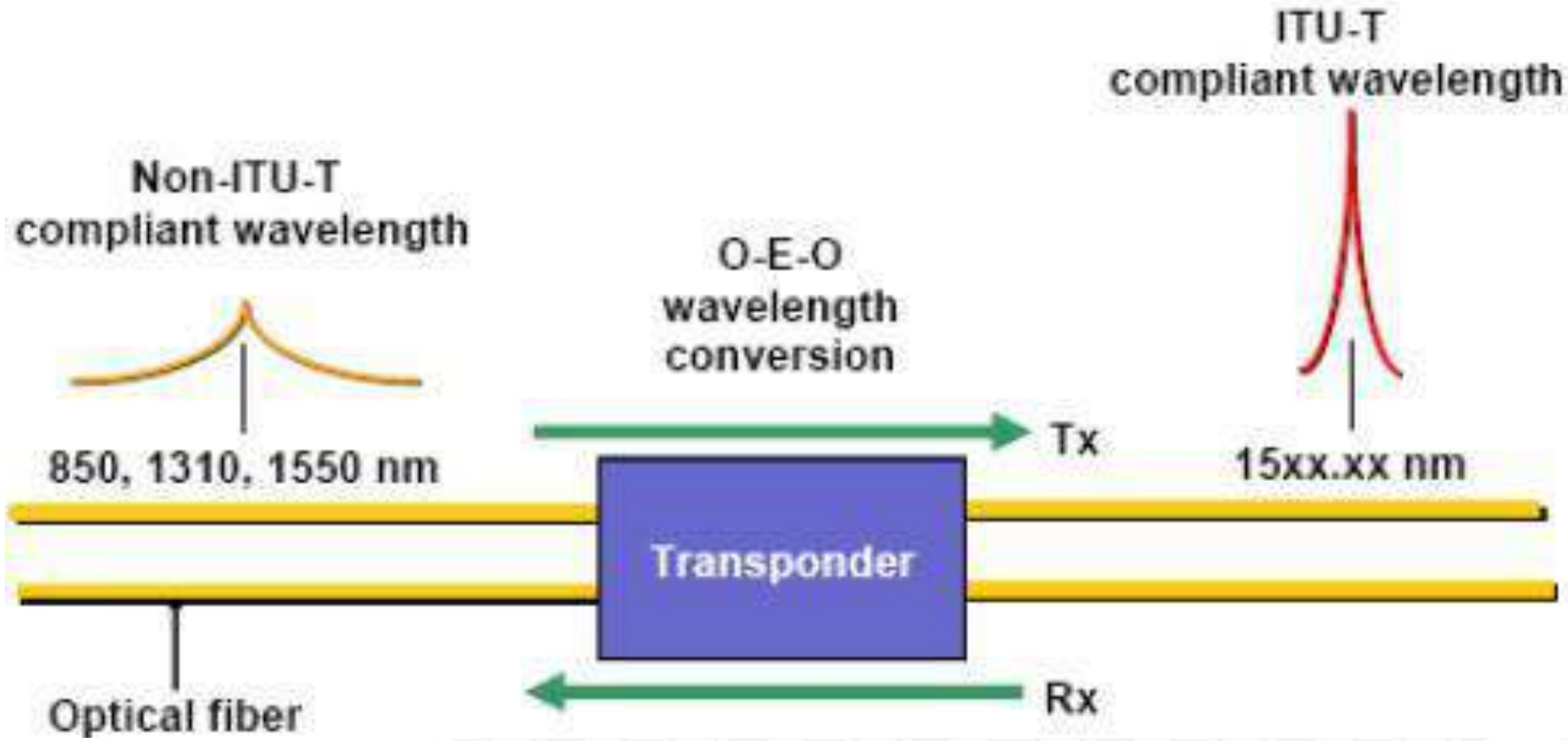
- Signal Conversion: It converts the client signal (which may be electrical or optical but at a non-DWDM wavelength) into a DWDM-compliant optical signal.
- Regeneration: In some cases, it also regenerates the signal to improve quality for long-distance transmission.
- Wavelength Assignment: It assigns a specific wavelength to each incoming signal.

Transponders (Cont.)

- Terminal Side (850/1310/1550 nm)
  The most common wavelengths on the terminal side are:
  - 850 nm: Used in multimode fiber systems, typically for short-distance connections.
  - 1310 nm: Common for single-mode fiber over medium distances (up to 10 km).
  - 1550 nm: Also used in single-mode fiber, especially for longer-distance connections (10-40 km).

- DWDM Side (15xx nm)
  Where the signals are converted into DWDM wavelengths for multiplexing onto a single optical fiber. DWDM systems operate in the C-band (1530-1565 nm) and sometimes in the L-band (1565-1625 nm) of the optical spectrum.
  These bands are chosen for their low attenuation and efficient transmission over long distances.

Transponders (Cont.)



Isolators and Circulators

Isolators

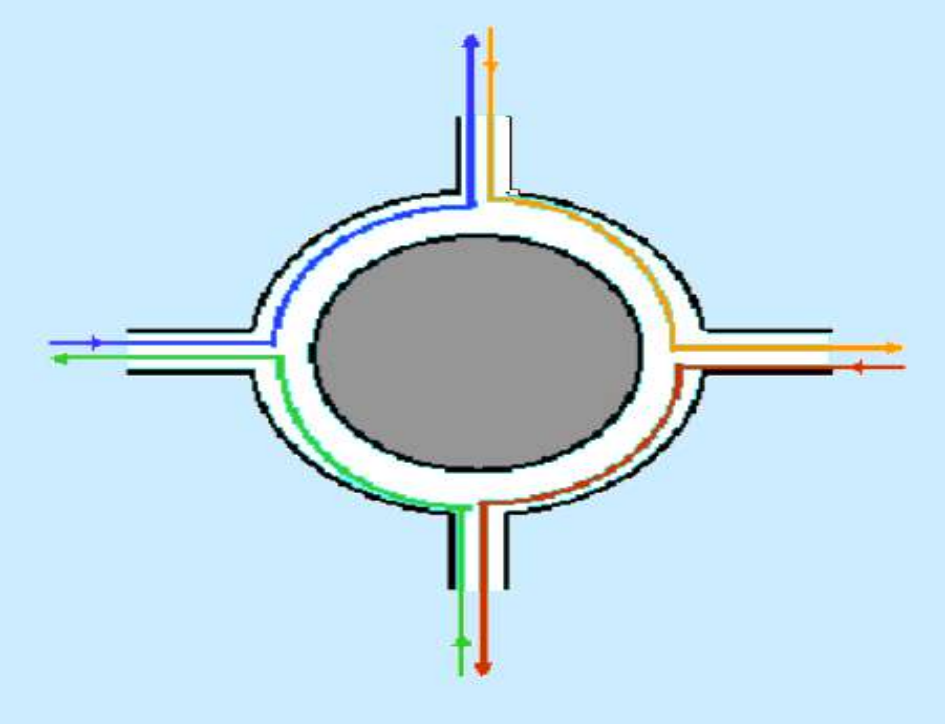
- allows light to pass in one direction but not in the opposite direction.

- used in the front and end of optical components such as amplifiers and laser diode modules.



Circulators

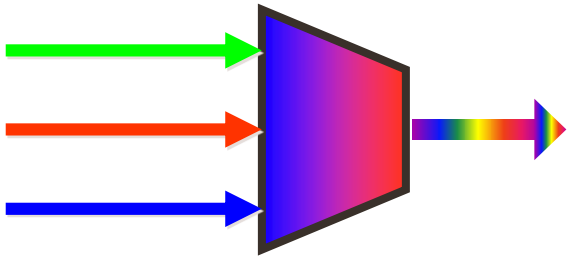
- Light entering at any particular port exits at the next port



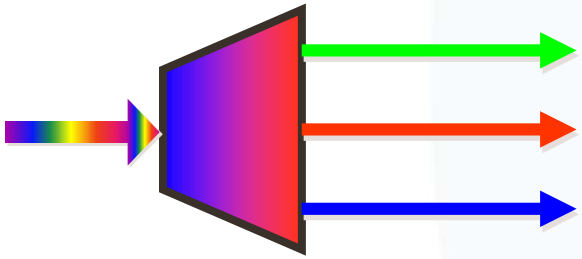


Multiplexers/Demultiplexers

DWDM Components (cont.)



Optical Multiplexer (MUX)



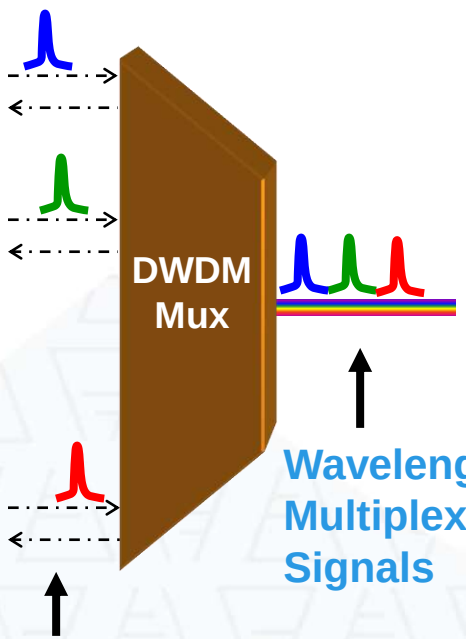
Optical De-multiplexer (DEMUX)

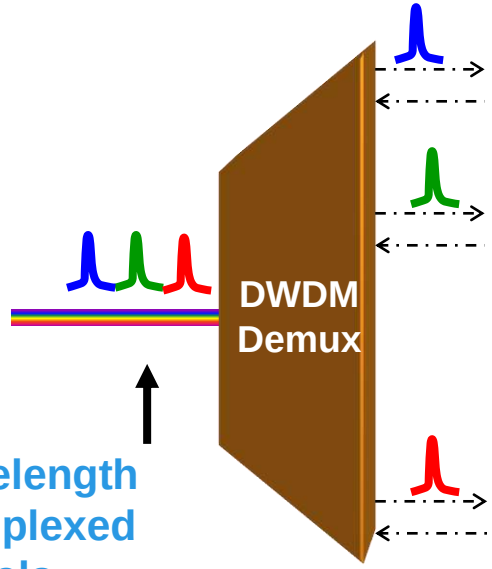
- **Multiplexer (Mux)**

Combines multiple optical signals (each with a different wavelength) into a single beam that consists of all these wavelengths.

- **Demultiplexer (Demux)**

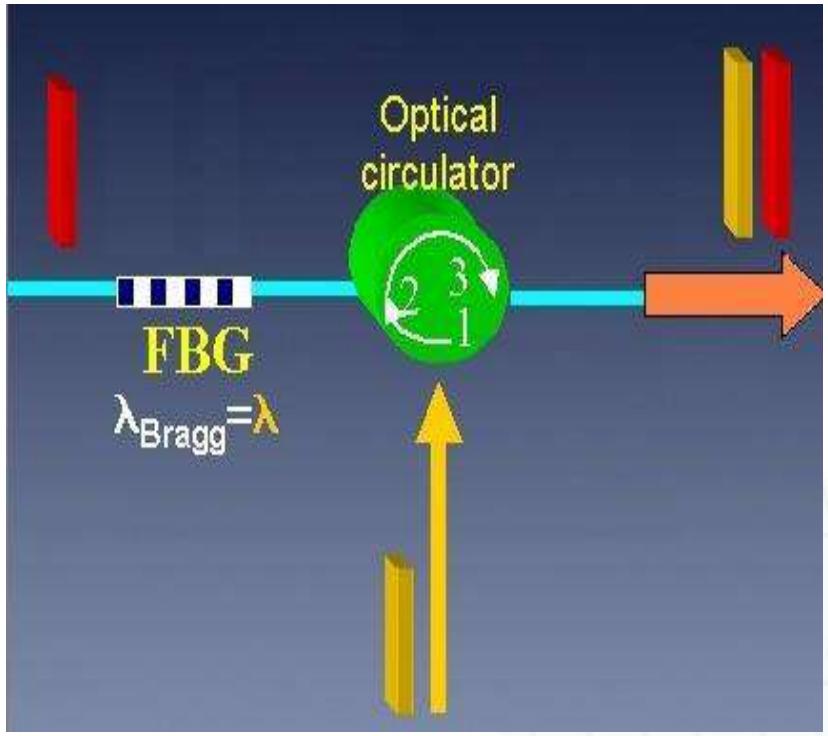
Performs the reverse functionality of multiplexers; Separates the combined optical signals at the receiver end into individual wavelengths, which are sent to the appropriate receiver.


DWDM
Mux
Wavelength
Multiplexed
Signals
Wavelengths
Converted via
Transponders

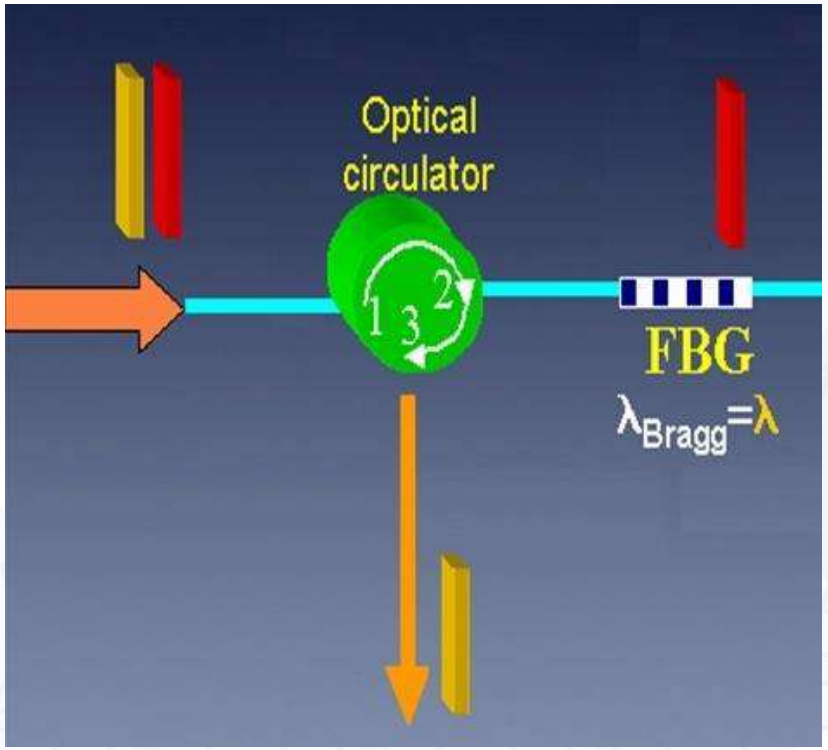

DWDM
Demux
Wavelength
Multiplexed
Signals
Wavelengths
separated into
individual ITU
Specific
lambdas

Circulator with FBG (De)Multiplexer

MUX



DeMUX

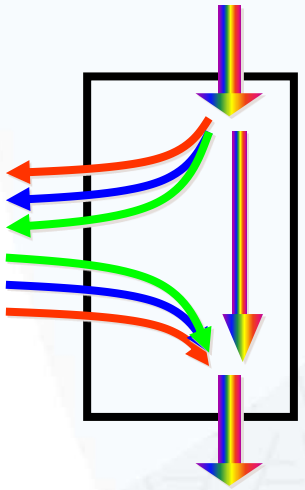


Add/Drop Multiplexers

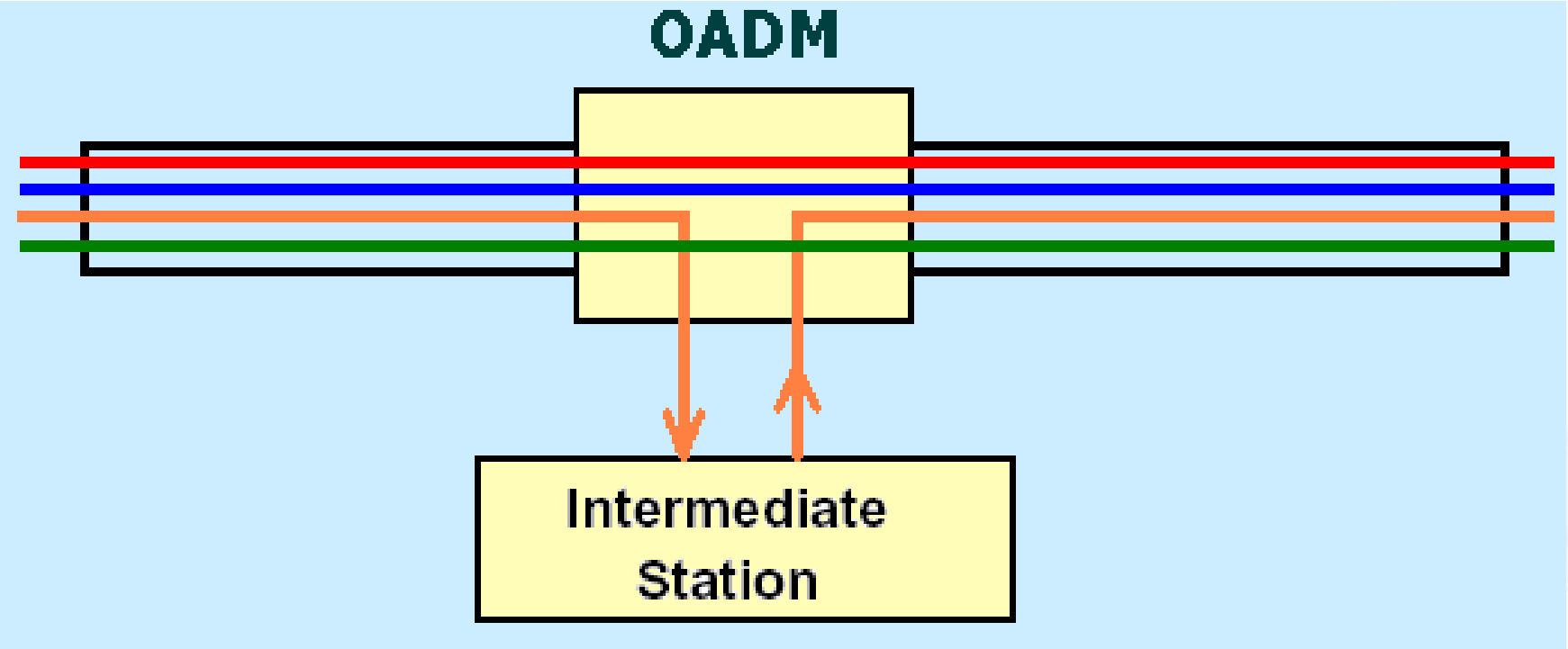
DWDM Components (cont.)

- **Optical Add-Drop Multiplexer (OADM)**

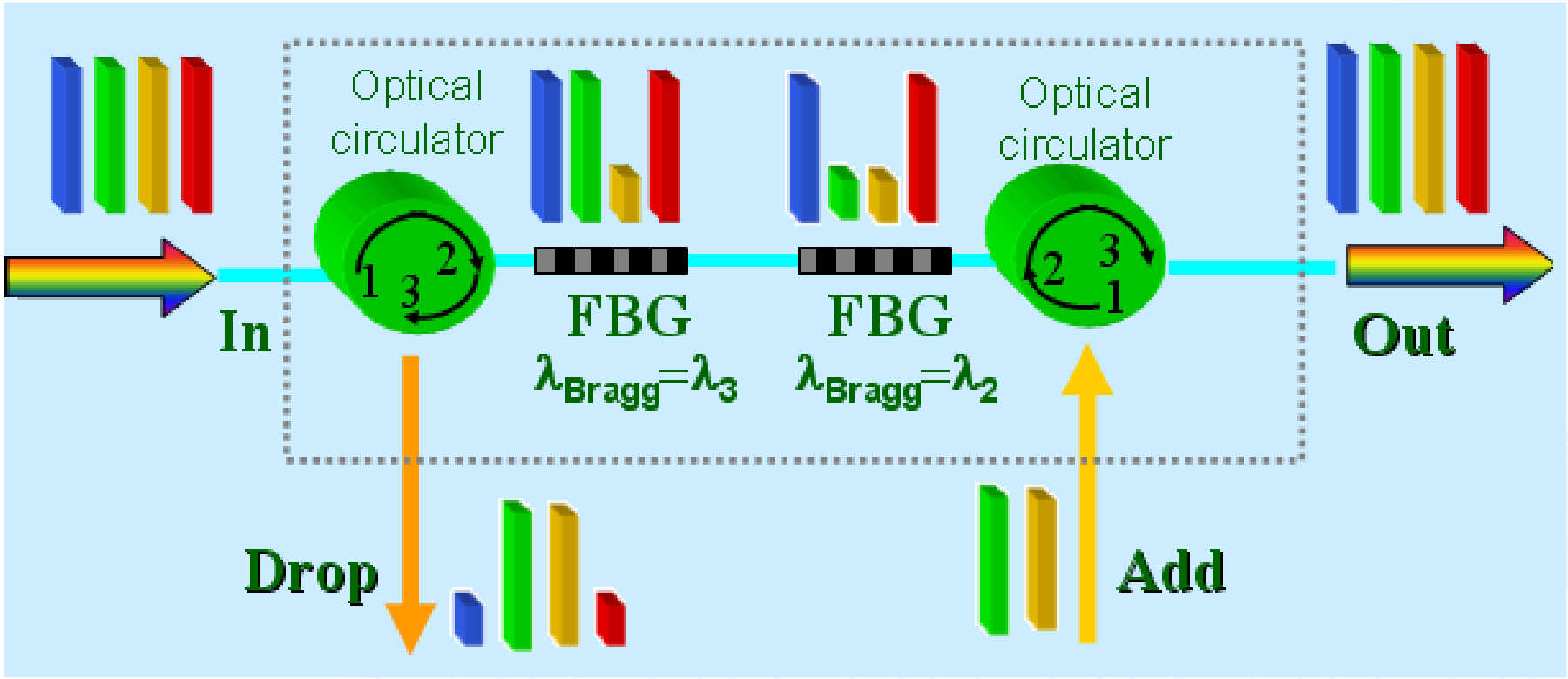
Adds or removes specific wavelengths from the DWDM signal while allowing other wavelengths to pass through. This is useful for routing specific channels to different destinations without disrupting the entire signal.



Optical Add/Drop Multiplexer (OADM)

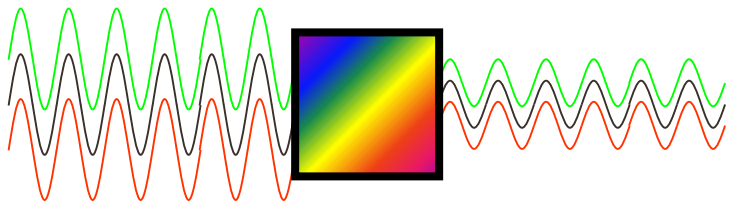

OADM
Intermediate
Station

Four channel OADM



Attenuators

DWDM Components (cont.)



Optical Attenuator
Variable Optical Attenuator
(VOA)

Attenuator

- Used to control the power level of the optical signal in a DWDM system.
- It adjusts the signal strength to prevent overloading of components (such as optical amplifiers or receivers) and ensures optimal signal quality across different wavelengths.
- Aid testing of optical system performance under varying optical power conditions.

Attenuators Types

1. Fixed Optical Attenuators

- **P**rovide a predetermined amount of attenuation.
- Commonly used when a specific level of signal reduction is needed, such as in testing or in long-haul communication links.

2. Variable Optical Attenuators (VOAs)

- **A**llow for adjustable attenuation levels.
- Used in applications where the required level of attenuation may change, such as in dynamic optical networks or during testing.



THANK YOU

QUESTIONS?





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